Lecture # 07

MM 318: Multicomponent Alloys

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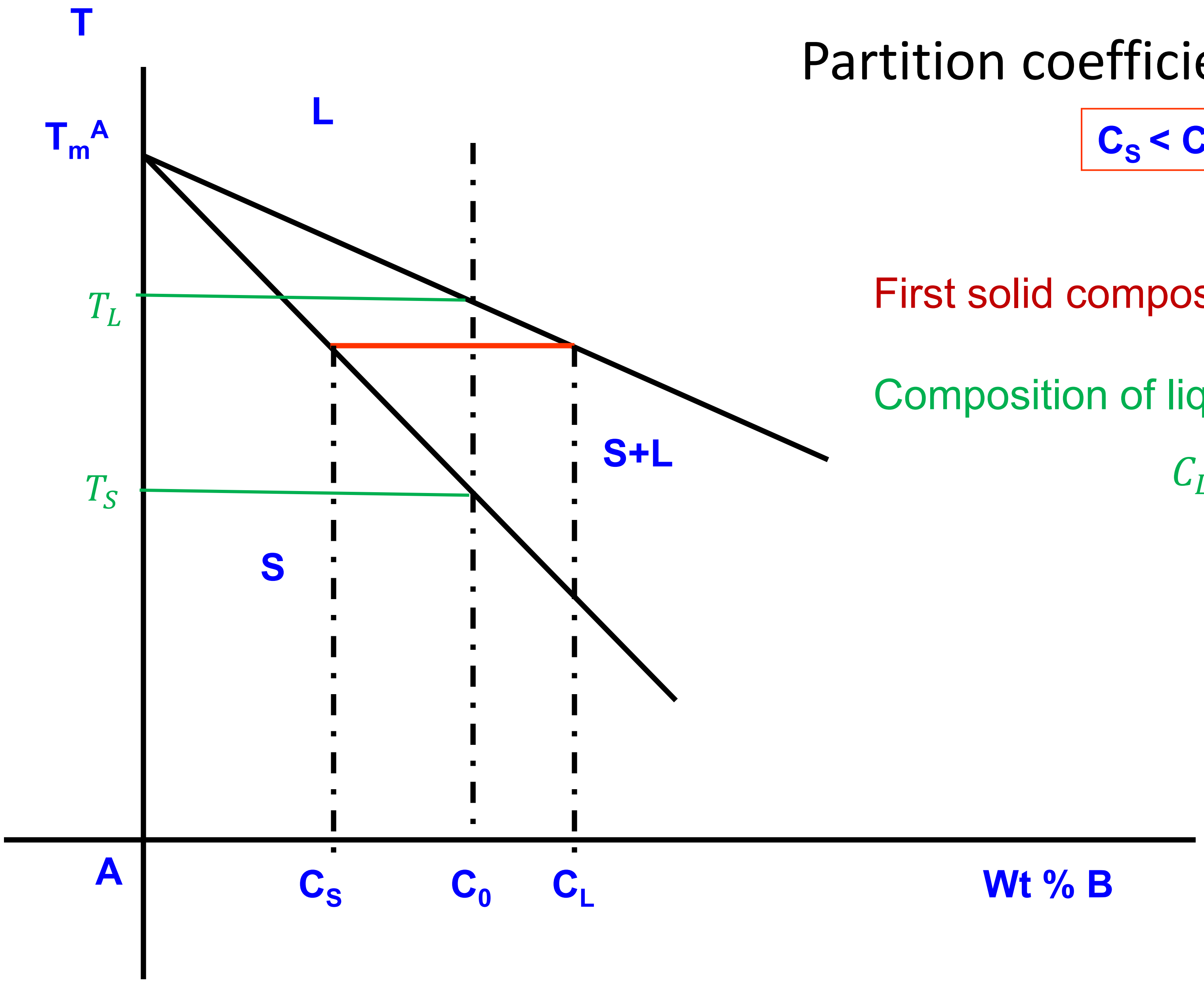
Mathematical analysis of solute redistribution during solidification

- Solidification at equilibrium and non-equilibrium condition

Partition coefficient: $k = \frac{C_S}{C_L}$

$\boxed{C_S < C_L}$



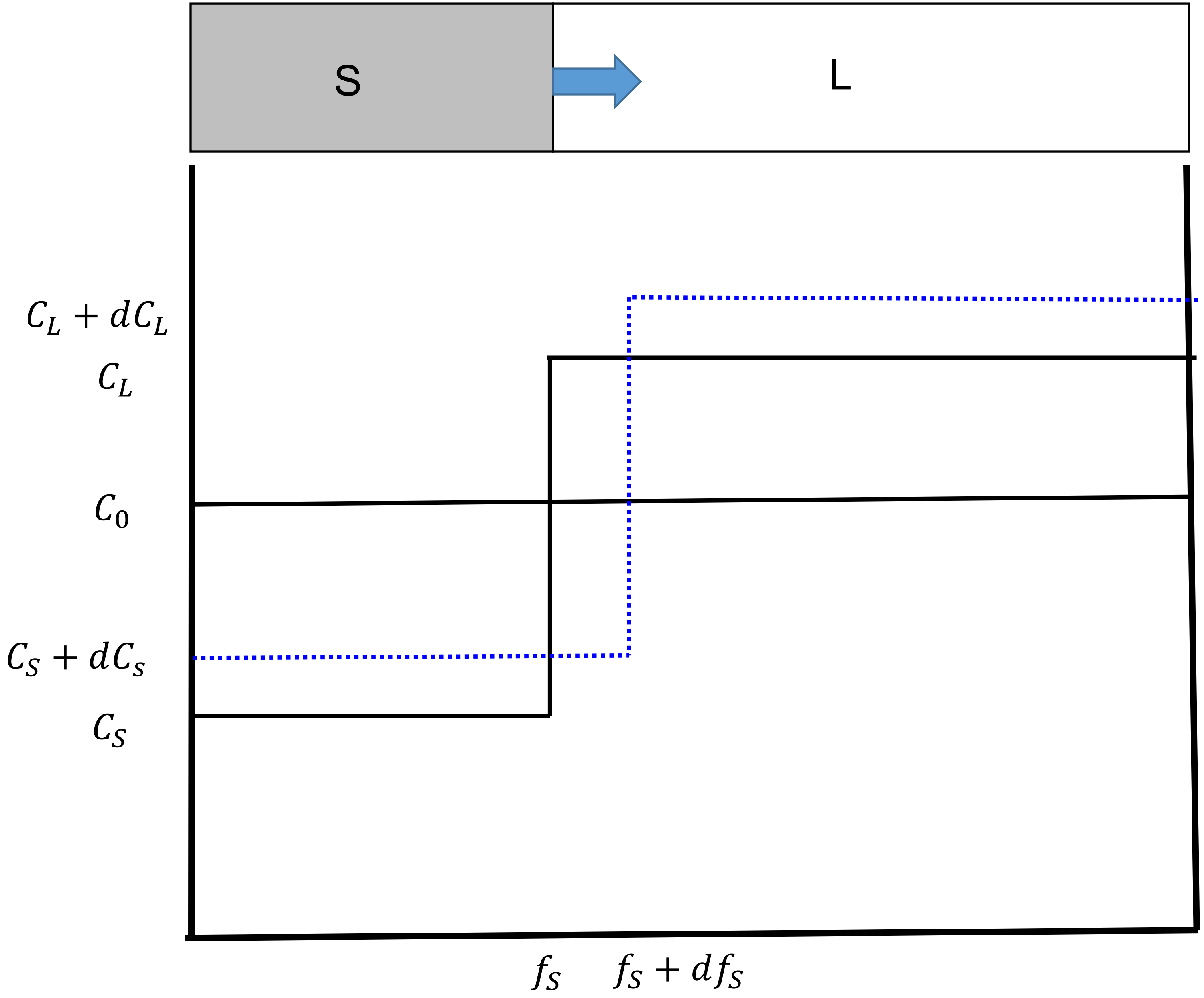
First solid composition: $C_S = kC_0$

Composition of liquid to be solidify last:

$$C_L = \frac{C_0}{k}$$

Portion of a binary phase diagram showing the equilibrium partition ratio

I. Complete mixing (complete diffusion in solid and liquid)



Solute balance:

Area under the solid line = Area under the dashed line

$$C_S f_S + C_L(1 - f_S)$$
$$= (C_S + dC_S)(f_S + df_S) + (C_L + dC_L)(1 - f_s - df_S)$$

I. Complete mixing (complete diffusion in solid and liquid)

Solute balance:

Area under the solid line = Area under the dashed line

$$C_S f_S + C_L(1 - f_S) = (C_S + dC_S)(f_S + df_S) + (C_L + dC_L)(1 - f_s - df_s)$$

$$\Rightarrow C_S\, f_S + C_L(1 - f_S) = C_S f_S + f_S dC_S + C_S df_S + \cancelto{0}{dC_S df_S} + C_L(1 - f_S) + dC_L(1 - f_S) - C_L df_S - \cancelto{0}{dC_L df_S}$$

$$\Rightarrow 0 = f_S dC_S + C_S df_S + dC_L(1 - f_S) - C_L df_S$$

$$\Rightarrow (C_L - C_S) df_S = f_S dC_S + dC_L(1 - f_S)$$

$$\Rightarrow (C_L - C_S) df_S = f_S dC_S + \frac{dC_S}{k}(1 - f_S)$$

$$(C_L - C_S)df_S = f_S dC_S + \frac{dC_S}{k}(1 - f_S)$$

$$(C_L - C_S)df_S = [\, f_S + \frac{(1 - f_S)}{k} dC_S \,]$$

$$\left(\frac{C_S}{k} - C_S\right) df_S = \left(\frac{kf_S + 1 - f_S}{k}\right) dC_S$$

$$C_S\left(\frac{1 - k}{k}\right) df_S = \left(\frac{1 - (1 - k)f_S}{k}\right) dC_S$$

$$\left(\frac{1 - k}{1 - (1 - k)f_S}\right) df_S = \frac{dC_S}{C_S}$$

On integrating both side;

$$(1 - k)\int_{f_s=0}^{f_s} \frac{df_S}{1 - (1 - k)f_S} = \int_{kC_0}^{C_S} \frac{dC_S}{C_S}$$

$$(1-k)\int_{f_s=0}^{f_s}\frac{df_S}{1-(1-k)f_S}=\int_{kC_0}^{C_S}\frac{dC_S}{C_S}$$

$$\Rightarrow (1-k)\ \frac{1}{(1-k)}\log\frac{1}{1-(1-k)f_s}=\log\frac{C_S}{kC_0}$$

$$\Rightarrow \log\frac{1}{1-(1-k)f_s}=\log\frac{C_S}{kC_0}$$

$$\Rightarrow \frac{1}{1-(1-k)f_s}=\frac{C_S}{kC_0}$$

$$\Rightarrow \frac{C_S}{k}=\frac{C_0}{1-(1-k)f_s}$$

$$\Rightarrow C_L=\frac{C_0}{1-(1-k)f_s}$$

$$C_L = \frac{C_0}{1-(1-k)f_s}$$

$$\Rightarrow C_L\,[1-(1-k)f_s] = C_0$$

$$\Rightarrow C_L - C_L(1-k)f_s = C_0$$

$$\Rightarrow C_L - (C_L - kC_L)f_s = C_0$$

$$\Rightarrow C_L - (C_L - C_S)f_s = C_0$$

$$\Rightarrow C_L - C_0 = (C_L - C_S)f_s$$

$$\Rightarrow f_s = \frac{C_L - C_0}{C_L - C_S}$$

$$f_L = 1 - f_s = 1 - \frac{C_L - C_0}{C_L - C_S} = \frac{C_0 - C_S}{C_L - C_S}$$

$$f_L = \frac{C_0 - C_S}{C_L - C_S}$$

Lever Rule:

Balance of solute under the assumption that complete mixing in solid and liquid.

Complete diffusion in both Solid and Liquid

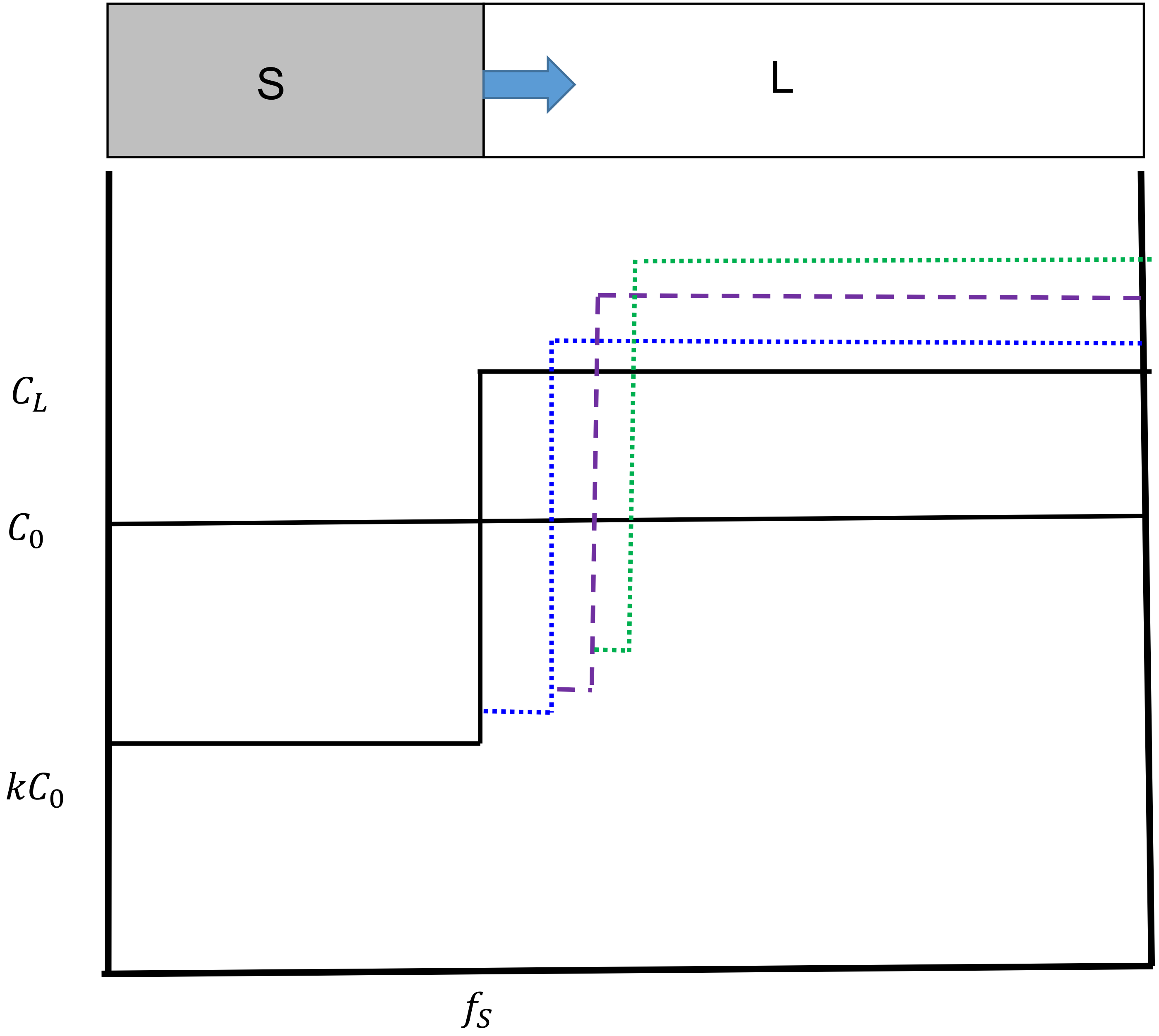
Lever Rule:

$$f_S = \frac{C_L - C_o}{C_L - C_S}$$

$$f_L = \frac{C_o - C_S}{C_L - C_S}$$

Solidification at equilibrium condition

II. Complete mixing in liquid, but no mixing in solid



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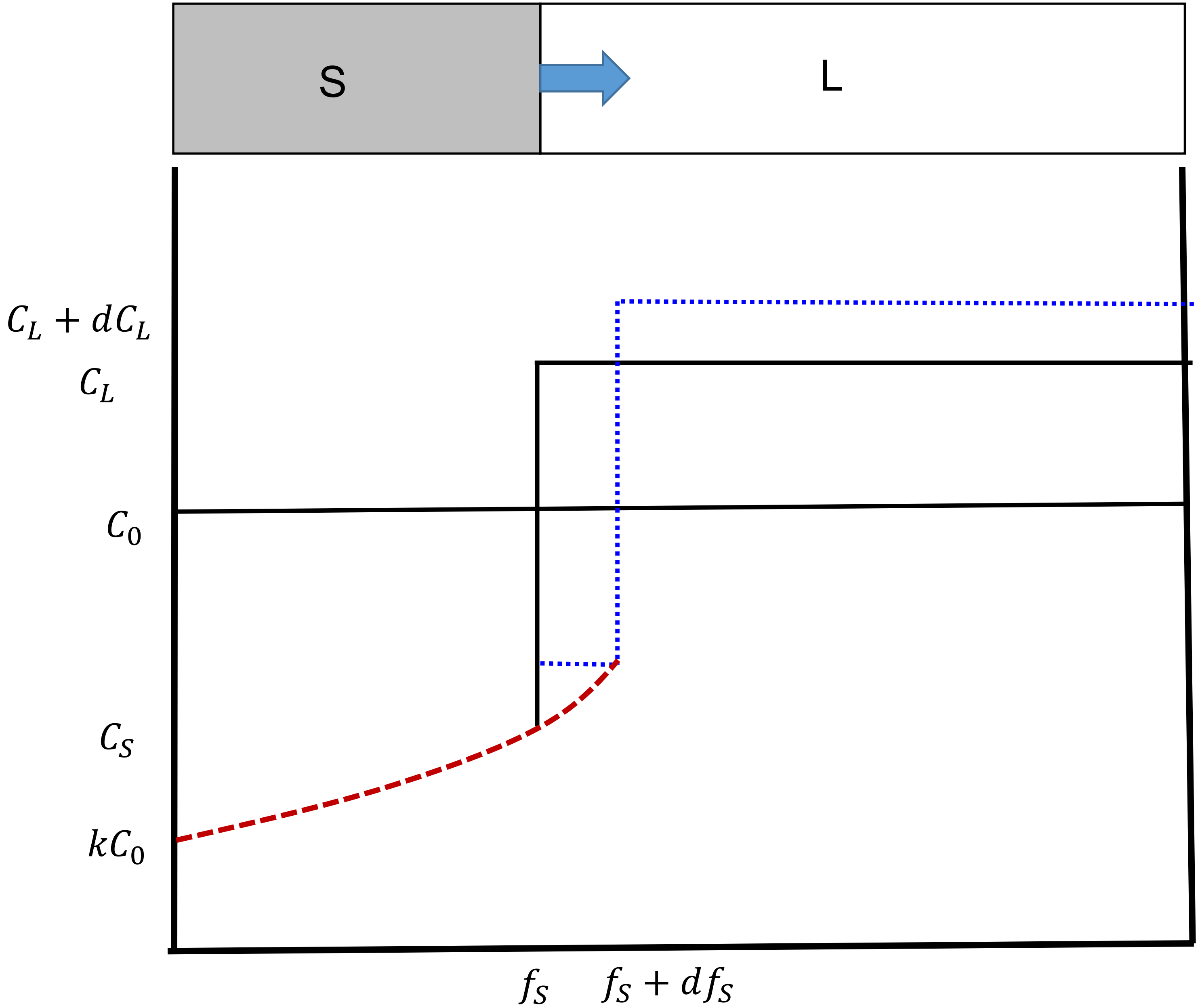


Solute balance:
Area under the solid line = Area under the dashed line

$$(C_L - C_S)df_S = (1 - f_S - df_S)dC_L$$

II. Complete mixing in liquid, but no mixing in solid

- Solute balance:

Area under the solid line = Area under the dashed line

$$(C_L - C_S)df_S = (1 - f_S - df_S)dC_L$$

$$\Rightarrow (C_L - C_S)df_S = (1 - f_S)dC_L - \cancelto{0}{df_S dC_L}$$

($df_S dC_L$ is small)

$$\Rightarrow (C_L - C_S)df_S = (1 - f_S)dC_L$$

$$\Rightarrow \frac{df_S}{(1 - f_S)} = \frac{dC_L}{(C_L - C_S)}$$

$$\Rightarrow \frac{df_S}{(1 - f_S)} = \frac{dC_L}{(C_L - kC_L)}$$

$$\Rightarrow \frac{df_S}{(1 - f_S)} = \frac{dC_L}{(1 - k)C_L}$$

$$\Rightarrow (1-k)\frac{df_S}{(1-f_S)} = \frac{dC_L}{C_L}$$

Integrating both sides, we get

$$\Rightarrow (1-k)\int_{f_S=0}^{f_S} \frac{df_S}{1-f_S} = \int_{C_0}^{C_L} \frac{dC_L}{C_L}$$

$$\Rightarrow (1-k)\log\left(\frac{1}{1-f_S}\right) = \log\left(\frac{C_L}{C_0}\right)$$

$$\Rightarrow \frac{C_L}{C_0} = \frac{1}{(1-f_S)^{1-k}}$$

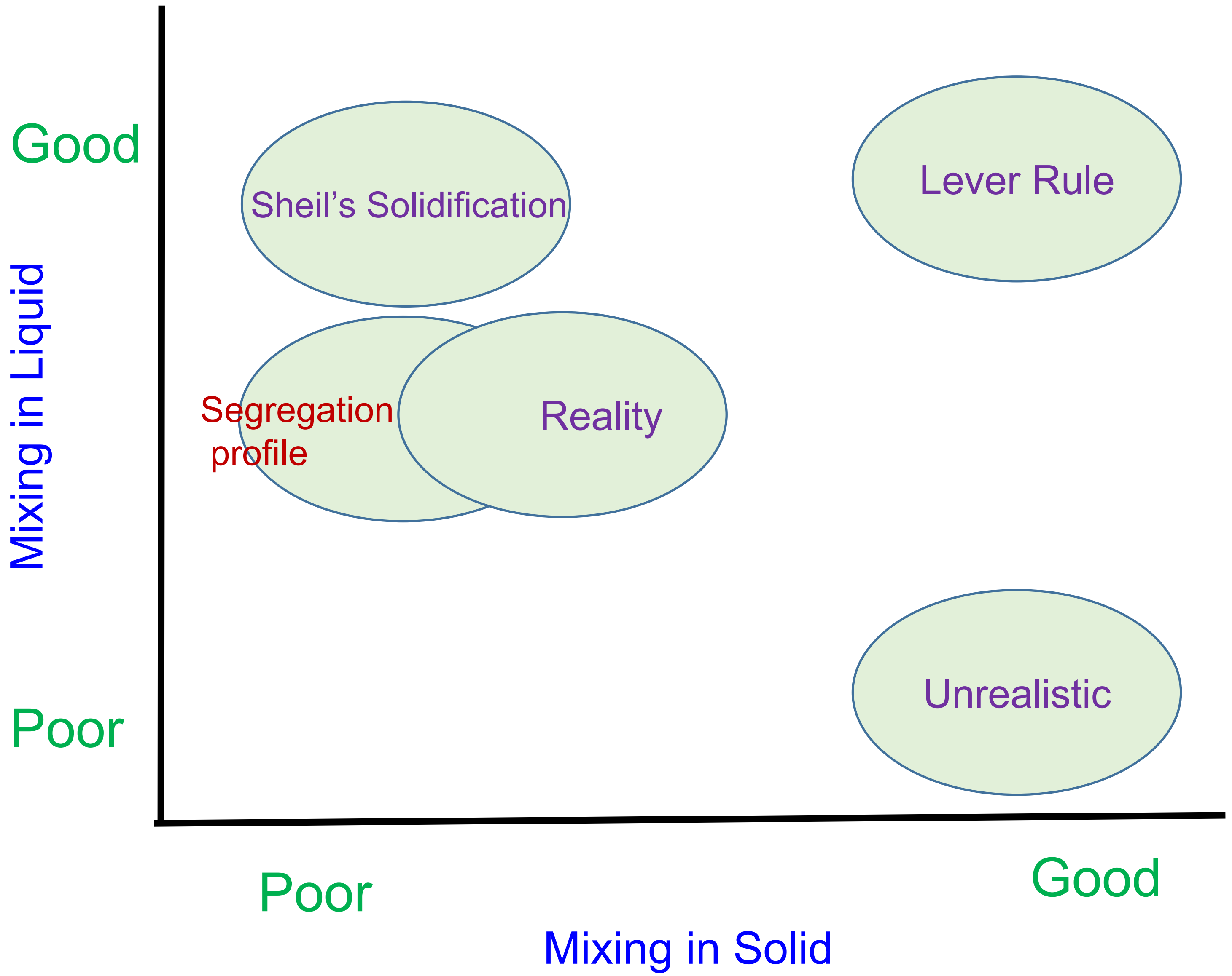
$$\Rightarrow C_L = \frac{C_0}{(1-f_S)^{1-k}}$$ (Scheil's Equation)

No diffusion in Solid, but complete diffusion in Liquid

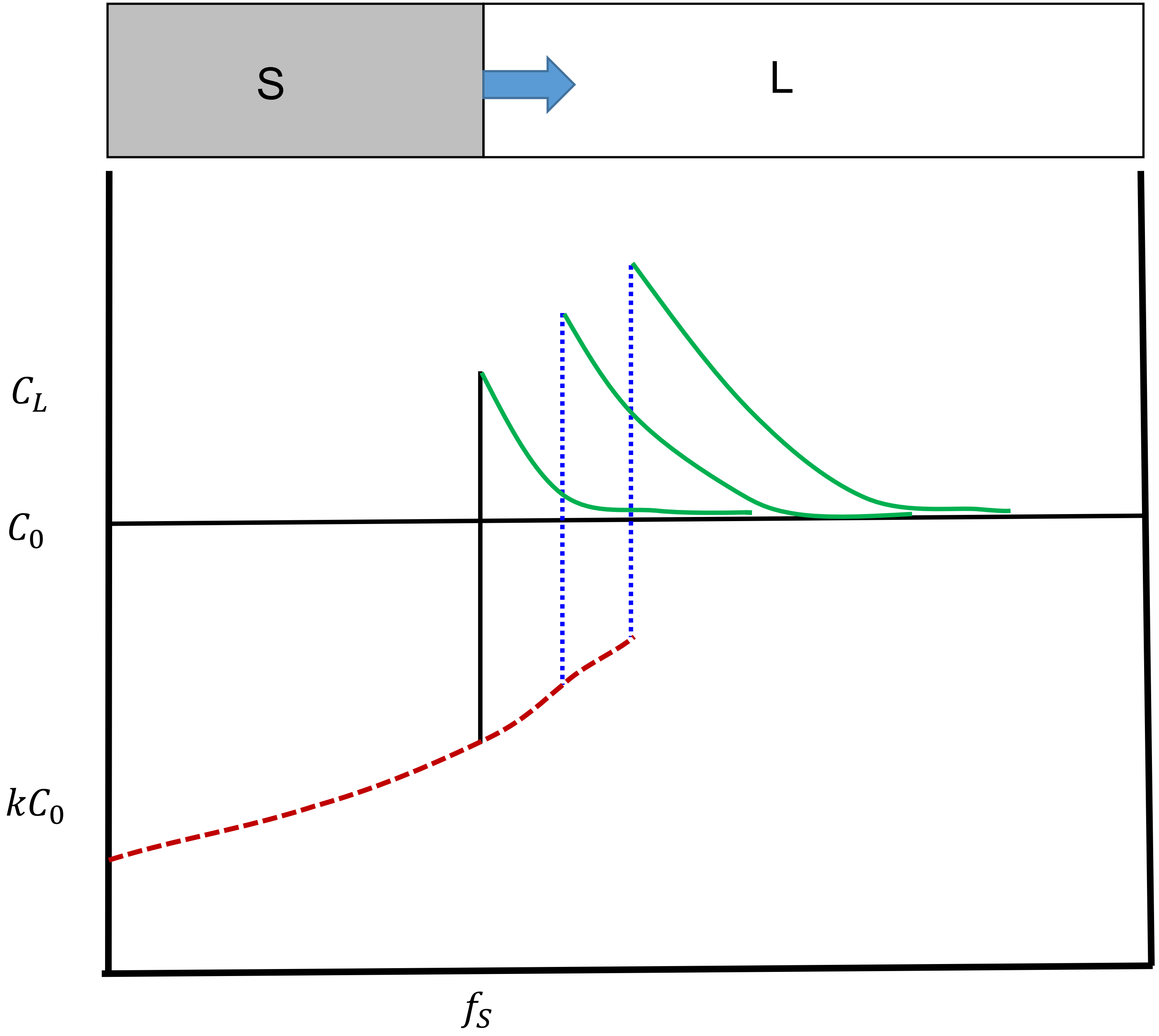
Scheils' Equation:

$$C_L = \frac{C_o}{(1 - f_s)^{1-k}}$$

Solidification at non-equilibrium condition


Good
Poor
Mixing in Liquid
Sheil's Solidification
Lever Rule
Segregation profile
Reality
Unrealistic
Poor
Good
Mixing in Solid

III. No solid diffusion and limited liquid diffusion



III. No solid diffusion and limited liquid diffusion

- **Problem statement:** 1D generalized Fick's law, subjected to initial conditions and boundary conditions

$$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$$

Boundary condition:

$$C_L\,(x=0) = \frac{C_0}{k}$$

$$C_L\,(x=\infty) = C_0$$

$$\frac{\partial}{\partial t} \rightarrow \frac{\partial}{\partial x}\frac{\partial x}{\partial t} \rightarrow v\,\frac{\partial}{\partial x}$$

$$-v\frac{\partial C}{\partial x} = D\,\frac{\partial^2 C}{\partial x^2}$$

Solute rejected by solidification = Solute taken away due to diffusive flux

$$\Rightarrow -v\,\dot{C} = D\,\ddot{C}$$

$$\Rightarrow\ \frac{\ddot{C}}{\dot{C}} = -\frac{v}{D}$$

Integrating w.r.t. once, we get

$$\log \dot{C} = -\frac{v}{D}\,x + Constant\ (A)$$

$$\log \dot{C} = -\frac{v}{D}\, x + A$$

$$\Rightarrow \dot{C} = A \exp(-\frac{v}{D}\, x)$$

Integrating one more time, we get

$$C = -\frac{D}{v}\, A \exp\left(-\frac{vx}{D}\right) + constant\ (B)$$

$$C = A' \exp\left(-\frac{vx}{D}\right) + B$$ (where $A' = -\frac{D}{v}\, A$)

At $\boldsymbol{x = 0; C_L = \frac{C_0}{k}}$

$$\boldsymbol{\frac{C_0}{k} = B + A'}$$

At $\boldsymbol{x} = \infty; \boldsymbol{C_L = C_0}$

$$C_0 = B$$

$$A' = C_0 \left(\frac{1-k}{k}\right)$$

$$C = C_L = A' \exp\left(-\frac{vx}{D}\right) + B$$

$$C_L = C_0 \left(\frac{1-k}{k}\right) \exp\left(-\frac{vx}{D}\right) + C_0$$

$$\Rightarrow \frac{C_L}{C_0} = 1 + \left(\frac{1-k}{k}\right) \exp\left(-\frac{vx}{D}\right)$$

No diffusion in Solid, but limited diffusion in Liquid

Segregation profile:

$$\frac{C_L}{C_o} = [1 + \frac{1-k}{k} e^{-vx/D}]$$

Solidification at non-equilibrium condition

Thank you